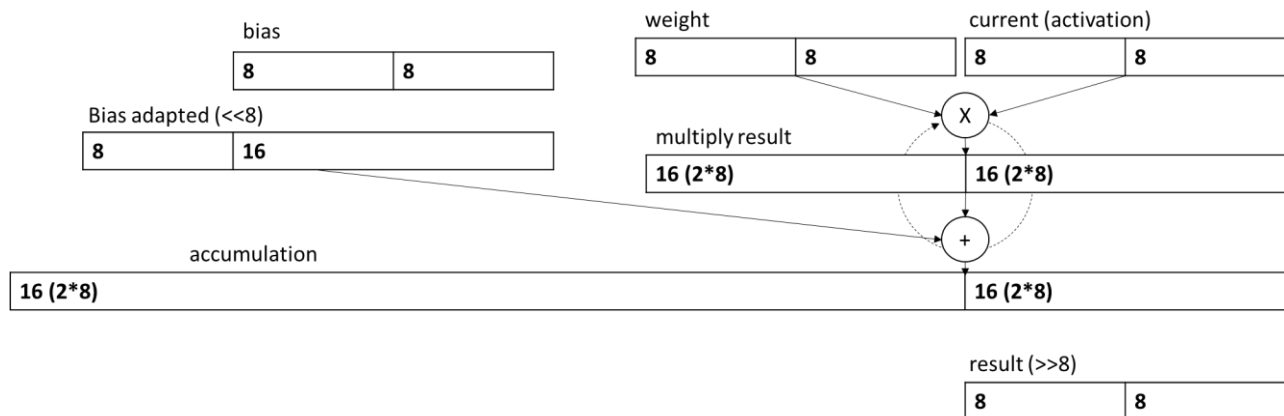


Tips and tricks for optimization

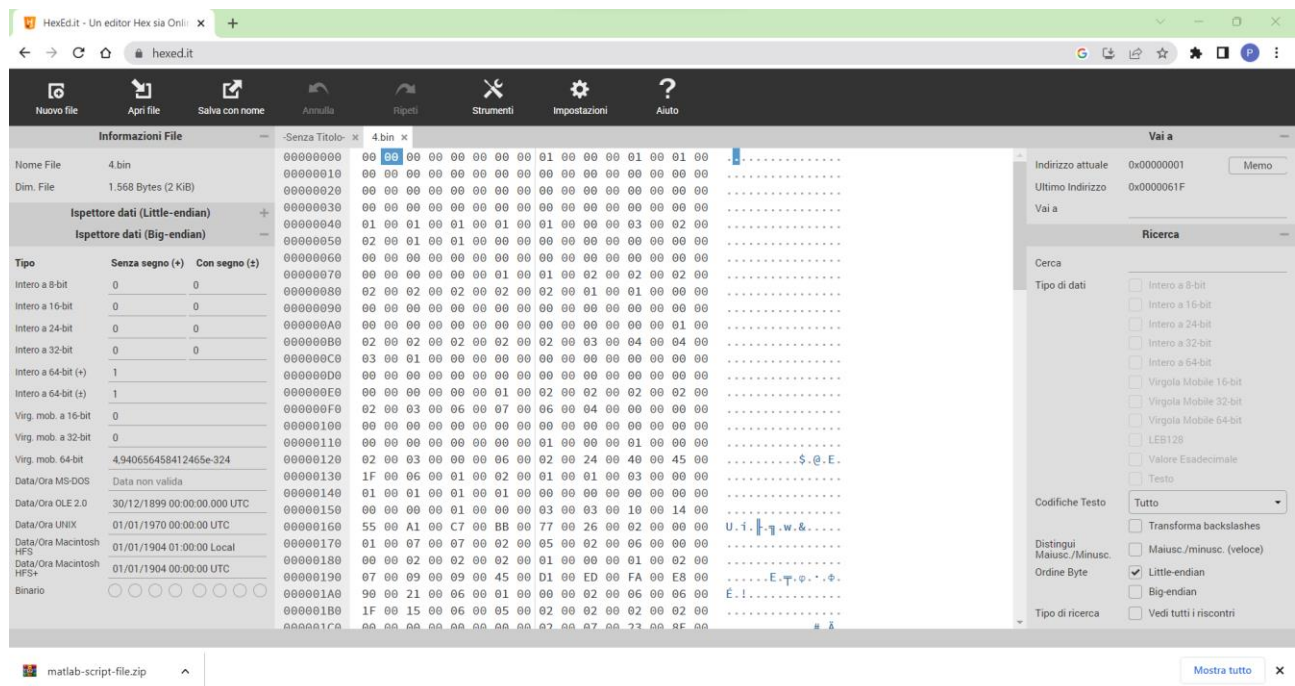
Clarification 1: The format selected for weights and activations is Q8.8

([https://en.wikipedia.org/wiki/Q_\(number_format\)](https://en.wikipedia.org/wiki/Q_(number_format)))

For example, check the implementation of the FC forward:



Tip 1: If you check using a hex editor the format of the image, you may notice that the most significant byte of the pixels (even column) is always 00.



All the pixels are positive.

Possible optimization: save time during the image send by creating a Q0.8 version of the input images and readapting pixels to Q8.8 on-board, after their receipt.

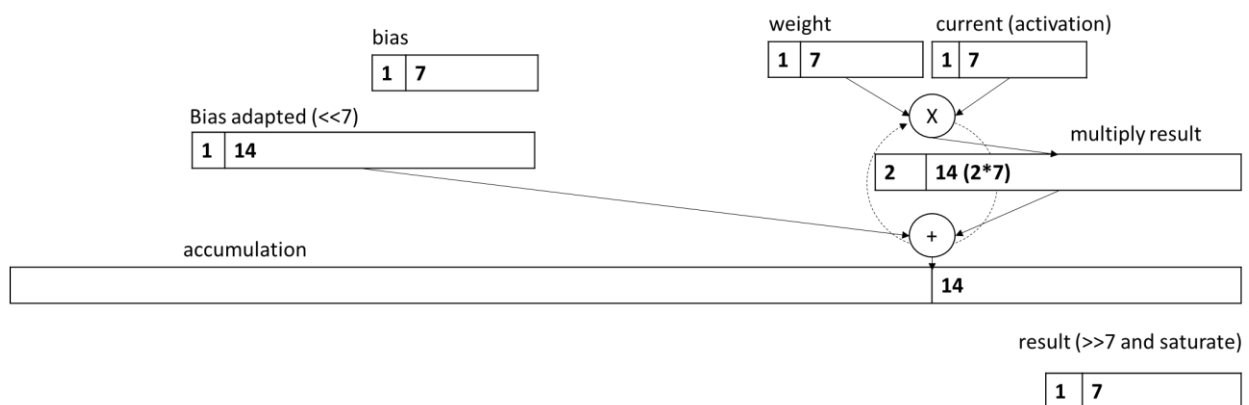
[To modify the files you could use any software of your choice, or you can use a setup similar to Lab2 (negative of an image over UART) to modify the pixels using the Zybo]

Tip 2: If you check weights and bias files using a hex editor, you may see that the most significant byte is always 0xFF or 0x00. Thus, weights are positive or negative but the integer part only represents the sign, and they can be represented with minimal impact using a Q1.7 format.

Possible improvements:

- Save time when sending weights via UART (modify weights and bias files to use the new format as you did for the image). Minimal gain since DNN parameters are only sent once at the startup
- Save memory occupation (modify weights and bias files to use the new format or save them in arrays using the new format when reading them from the UART)

Possible improvement (advanced): Try to use 8 bits for both weights, bias and activations to exploit better SIMD. Using reduced precision for both weights, bias and activations would require to adapt all to the same QF. Use saturate() to adapt results of the FC forward to the new data format.



Exploration of the optimal QF could be needed.